

HDI Predictor - Final Technical Report

Abstract

The Human Development Index (HDI) serves as a primary metric for assessing national growth. However, delayed data publishing creates challenges. This project details 'HDI Insight AI' - a web analytics platform that provides real-time classifications of developmental categories using a Support Vector Machine (F1-score: 98.9%) and Random Forest score regressor.

Introduction & Objectives

Traditional human development monitoring cycles are static. This platform solves the delay by deploying an interactive, responsive web simulator. Policymakers can adjust parameters and receive explanations and recommendations instantly.

Algorithms & Results

Benchmarked algorithms (Logistic Regression, Decision Trees, Random Forests, SVM, KNN). SVM outperformed other models, achieving 98.9% accuracy. The model was serialized to disk and loaded into Flask for server-side inferences.

Conclusion

We successfully designed and deployed a portfolio-ready machine learning application following professional SDLC practices, integrating analytics, Explainable AI, and reporting engines into a clean, modern user interface.